Name: _____.

Student number: _____.

Instructions

- Turn off all the electronic devices.
- This is a closed book, and no notes exam.
- The last page of this exam contains a notesheet that you can tear off.
- You can use any fact in the notesheet, but you must reference which fact you are using.
- You can use every theorem proved in class, but you need to state the theorem.
- Unless otherwise stated, you must justify your answers.
- If you need more room, use the backs of the pages and indicate that you have done so.
- If you have a question, raise your hand and I will come to you.
- You have 50 minutes to complete the exam.
- Good luck!

Question:	1	2	3	4	5	Total
Points:	10	10	10	10	10	50
Score:						

1. (10 points) For each of the following statements, write the negation of the statement without using the logical operator "$\neg$" or the word "not." You can use either symbolic logic or an English sentence. Determine whether the original statement or its negation is true, and give a proof of the true statement.

   (a) $(\forall n \in \mathbb{Z})(\exists m \in \mathbb{Z})(n + m = -n)$.

Negation: $(\exists n \in \mathbb{Z})(\forall m \in \mathbb{Z})(n+m \neq -n)$

The original statement is true.

In fact, given $n \in \mathbb{Z}$, we can take $m = -2 \cdot n$ which is also in $\mathbb{Z}$ because $-2 \in \mathbb{Z}$ and $n \in \mathbb{Z}$, and which satisfies

$n + m = n + (-2)n = (1-2)n = -n$, as desired.

   (b) There exists a real number x such that, for any real number y, if $x + y > 0$ then $y > 0$.

The negation is "for every real number x, there exists a real number y such that $x + y > 0$ and $y \leq 0$".

The original statement is true.

To prove that, we just need to find a particular $x \in \mathbb{R}$ such that, for each $y \in \mathbb{R}$ with $x + y > 0$, we have $y > 0$.

Take $x = 0$. Assume $y \in \mathbb{R}$ is such that $x + y > 0$. Since $x + y = 0 + y = y$, we also have $y > 0$, as desired.

2. (10 points) Let P, Q and R be statements. Show that the following logical equivalences hold.

   (a) $\big((P \to Q) \wedge (R \to Q)\big) \equiv \big((P \vee R) \to Q\big)$.

$(P \to Q) \wedge (R \to Q) \equiv (\neg P \vee Q) \wedge (\neg R \vee Q) \equiv (\neg P \wedge \neg Q) \vee R$

$S \to T \equiv \neg S \vee T$ Twice

Distributive

$\equiv \neg(P \vee Q) \vee R$

De Morgan

$\equiv (P \vee Q) \to R$

$S \to T \equiv \neg S \vee T$

   (b) $\neg(P \leftrightarrow Q) \equiv (P \wedge \neg Q) \vee (Q \wedge \neg P)$.

$\neg(P \leftrightarrow Q) \equiv \neg\big((P \to Q) \wedge (Q \to P)\big) \equiv \neg(P \to Q) \vee \neg(Q \to P)$

De Morgan

$\equiv \neg(\neg P \vee Q) \vee \neg(\neg Q \vee P)$

Twice $S \to T \equiv \neg S \vee T$

$\equiv (\neg\neg P \wedge \neg Q) \vee (\neg\neg Q \wedge \neg P)$

De Morgan Twice

$\equiv (P \wedge \neg Q) \vee (Q \wedge \neg P)$

3. (10 points) Decide whether the following statements are true or false. If the statement is true, prove it. If it is false, give a counterexample.

   (a) Let A, B and C be sets. Then $(A \setminus B) \cap (A \setminus C) = A \setminus (B \cap C)$.

This is False.

Let $A = \{1, 2\}$, $B = \{1\}$ and $C = \{2\}$

Then $(A \setminus B) \cap (A \setminus C) = \{2\} \cap \{1\} = \emptyset$

But $A \setminus (B \cap C) = A \setminus \emptyset = A = \{1, 2\}$

Since A is non-empty, we have

$$(A \setminus B) \cap (A \setminus C) \neq A \setminus (B \cap C)$$

   (b) The set $A = \{7m \mid m \in \mathbb{Z}\}$ is closed under addition.

This is True. We need to show that for any $x \in A$ and $y \in A$, we have also $x + y \in A$.

If $x \in A$ and $y \in A$, we have $x = 7m$ and $y = 7n$ for some $m, n \in \mathbb{Z}$. But then

$$x + y = 7m + 7n = 7(m + n)$$

By integer closure, $m + n \in \mathbb{Z}$, so this shows that $x + y \in A$, as desired.

4. (10 points) Show that $3^n > 2^n + 1$ for every integer $n \geqslant 2$.

Proof: By induction on $n \geq 2$.

Base case: For $n=2$, We have $3^n = 3^2 = 9$ and $2^n+1 = 2^2+1 = 5$,

So $3^n > 2^n + 1$ for $n=2$.

Inductive step: Assume That $3^k > 2^k + 1$ for some $k \geq 2$.

We want to show That $3^{k+1} > 2^{k+1} + 1$. We compute

$$3^{k+1} = 3^k \cdot 3 > (2^k+1)\,3 = 3\cdot 2^k + 3 > 2\cdot 2^k + 3 = 2^{k+1} + 3 > 2^{k+1} + 1$$

- $3^k > 2^k + 1$ by Ind Hyp and $3 > 0$
- $3 > 2$ and $2^k > 0$ Plus fact R12
- $3 > 1$ and R12 again

Thus $3^{k+1} > 2^{k+1} + 1$, as desired

It follows by induction That $3^n > 2^n + 1$ for every $n \geq 2$ □

5. (10 points) Recall that the Fibonacci numbers are defined by $F_0 = 1$, $F_1 = 1$ and $F_n = F_{n-1} + F_{n-2}$ for $n \geqslant 2$.

   Prove that $\sum_{i=0}^{n} F_i = F_{n+2} - 1$ for every integer $n \geqslant 0$.

We prove this by induction on $n \geq 0$.

Base case: For $n=0$, we have $\sum_{i=0}^{n} F_i = \sum_{i=0}^{0} F_i = F_0 = 1$

and also $F_{n+2} - 1 = F_2 - 1 = 2 - 1 = 1$, so the base case holds.

Inductive step: Assume $\sum_{i=0}^{k} F_i = F_{k+2} - 1$ (Ind Hyp) for some $k \geq 0$.

We want to show $\sum_{i=0}^{k+1} F_i = F_{k+3} - 1$. We compute LHS:

$$\sum_{i=0}^{k+1} F_i = \left(\sum_{i=0}^{k} F_i\right) + F_{k+1} = F_{k+2} - 1 + F_{k+1}$$

(Ind Hyp)

$$= F_{k+1} + F_{k+2} - 1$$

This works because $k+3 \geq 2$ $\forall k \geq 0$.

$$= F_{k+3} - 1, \quad \text{as desired}$$

It follows by induction that $\sum_{i=0}^{n} F_i = F_{n+2} - 1$ for all $n \geq 0$. □

Axioms

Field axioms

1. **Closure:** If $a, b \in \mathbb{R}$, then $a + b \in \mathbb{R}$ and $ab \in \mathbb{R}$.
2. **Commutativity:** If $a, b \in \mathbb{R}$, then $a + b = b + a$ and $ab = ba$.
3. **Associativity:** If $a, b, c \in \mathbb{R}$, then $(a + b) + c = a + (b + c)$ and $(ab)c = a(bc)$.
4. **Distributivity:** If $a, b, c \in \mathbb{R}$, then $(a+b)c = ac+bc$ and $a(b + c) = ab + ac$.
5. **Existence of Identities:** There are two different real numbers 0 and 1 such that for all real numbers a, $a + 0 = a$ and $a \cdot 1 = a$.
6. **Additive Inverses:** For all $a \in \mathbb{R}$, there exists $-a \in \mathbb{R}$ such that $a + (-a) = 0$.
7. **Multiplicative Inverses:** For all $a \in \mathbb{R}$ such that $a \neq 0$, there exists $a^{-1} \in \mathbb{R}$ such that $aa^{-1} = 1$.

- $\mathbb{N} \cup \{0\}$ satisfies Axioms 1–5 (quantified over $\mathbb{N}$)
- $\mathbb{Z}$ satisfies Axioms 1–6 (quantified over $\mathbb{Z}$)
- $\mathbb{Q}$ satisfies Axioms 1–7 (quantified over $\mathbb{Q}$)

Equality Axioms. These hold for $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$.

8. **Reflexivity:** For every real number a, $a = a$.
9. **Symmetry:** For all real numbers a and b, if $a = b$, then $b = a$.
10. **Transitivity:** For all real numbers a, b, and c, if $a = b$ and $b = c$ then $a = c$.
11. **Substitution:** If $a = b$, then b may be substituted for a in any statement.

Order Axioms. We assume $\mathbb{R}$ contains a subset $\mathbb{R}^+$ of "positive real numbers" such that

12. **Positive Closure:** If $a, b \in \mathbb{R}^+$, then $a + b \in \mathbb{R}^+$ and $ab \in \mathbb{R}^+$.
13. **Trichotomy:** If $a \in \mathbb{R}$, then exactly one of the following is true:

    • $a \in \mathbb{R}^+$ • $-a \in \mathbb{R}^+$ • $a = 0$.

14. **Well-Ordering Property:** If S is a nonempty set of positive integers, then S has a minimum element.

Facts

Logic. Let P, Q and R be statements.

- $P \to Q \equiv \neg P \vee Q$
- $P \leftrightarrow Q \equiv (P \to Q) \wedge (Q \to P)$

De Morgan's Laws

- $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$
- $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$

Distributive Laws

- $P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$
- $P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$

Sets. Let A and B be subsets of universe U.

- $A \subseteq B$ if $x \in A$ implies $x \in B$.
- $A = B$ if $A \subseteq B$ and $B \subseteq A$.
- $A \cup B = \{x \in U : x \in A \text{ or } x \in B\}$
- $A \cap B = \{x \in U : x \in A \text{ and } x \in B\}$
- $A \backslash B = \{x \in U : x \in A \text{ and } x \notin B\}$
- $A^c = \{x \in U : x \notin A\}$

Inequalities. Let $a, b \in \mathbb{R}$.

- We say $a < b$ if $b - a \in \mathbb{R}^+$.
- We say $a \leqslant b$ if $a < b$ or $a = b$.
- We say $a > b$ if $a - b \in \mathbb{R}^+$.
- We say $a \geqslant b$ if $a > b$ or $a = b$.

Real numbers. Let $a, b, c \in \mathbb{R}$.

R1 If $a = b$ then $a + c = b + c$ and $ac = bc$.
R2 If $a + c = b + c$ then $a = b$.
R3 The additive identity in $\mathbb{R}$ is unique.
R4 $0 \cdot a = 0$.
R5 $(-1)a = -a$.
R6 $(-1)(-1) = 1$
R7 $(-a)(-b) = ab$
R8 $-(ab) = (-a)b = a(-b)$.
R9 1 is positive.
R10 $a \in \mathbb{R}^+$ if and only if $a > 0$.
R11 if $a > b$ and $c > 0$ then $ac > bc$.
R12 if $a > b$ then $a + c > b + c$.